



United International University
Department of Computer Science and Engineering
Course Code: CSE 2215 Course Title: Data Structure and
Algorithms I (B)

Class Test – 2
Duration: 30mins (all inclusive)

Date: 07/08/2021
Marks:20

1. Suppose there is a singly linked node declared as below: [10]

```
struct node{  
    int value;  
    struct node *next;  
}
```

And the linked list looks like:

Head → 6 → 8 → 7 → 9 → 5 → NULL

Develop a technique to remove the node with the highest value. Explain your technique using diagrams. Draw separate diagrams for every single step. You do not need to write any code or pseudocode.

2. Using the values given below after your student id as the array, partition the array. Use the last index value as pivot. Draw an array for each step of your work. [10]

Student ID	Values
011173044	[17, 25, 94, 80, 34, 11, 84, 86, 38, 49]
011181265	[78, 75, 84, 86, 15, 50, 12, 17, 57, 96]
011183017	[77, 68, 67, 93, 71, 95, 10, 59, 74, 61]
011183020	[72, 60, 89, 22, 57, 24, 15, 88, 70, 59]
011183056	[85, 58, 60, 34, 98, 59, 67, 41, 77, 71]
011183077	[14, 83, 82, 66, 25, 27, 76, 10, 61, 62]
011191160	[14, 48, 79, 57, 19, 50, 77, 45, 18, 44]
011191173	[62, 55, 24, 94, 78, 47, 37, 89, 88, 20]
011191179	[65, 44, 93, 91, 97, 13, 32, 77, 51, 24]
011191212	[25, 53, 66, 39, 54, 13, 49, 50, 80, 69]
011192083	[81, 85, 77, 29, 20, 24, 82, 18, 83, 25]
011192094	[74, 52, 85, 27, 71, 81, 35, 97, 98, 31]
011192122	[11, 10, 61, 42, 69, 67, 23, 47, 12, 39]

011193028	[59, 24, 78, 91, 60, 42, 35, 90, 37, 69]
011193038	[66, 16, 65, 37, 63, 91, 42, 67, 74, 32]
011193040	[96, 26, 45, 33, 64, 53, 32, 55, 52, 71]
011193045	[27, 75, 91, 10, 61, 83, 39, 25, 70, 49]
011193047	[19, 70, 85, 44, 36, 50, 13, 14, 97, 57]
011193052	[75, 92, 61, 65, 13, 79, 58, 42, 44, 85]
011193068	[41, 33, 81, 98, 86, 48, 11, 31, 69, 65]
011193122	[60, 51, 12, 62, 81, 54, 88, 27, 80, 48]
011201019	[66, 18, 65, 40, 98, 27, 58, 57, 89, 12]
011201031	[59, 40, 38, 22, 24, 61, 85, 91, 14, 37]
011201037	[26, 19, 39, 55, 49, 30, 75, 27, 54, 63]
011201084	[69, 58, 62, 87, 90, 25, 12, 92, 15, 19]
011201088	[28, 94, 47, 63, 86, 45, 14, 76, 12, 56]
011201108	[85, 46, 70, 23, 14, 66, 86, 72, 44, 62]
011201125	[18, 49, 90, 53, 47, 44, 76, 22, 88, 98]
011201128	[58, 95, 72, 66, 52, 42, 93, 91, 24, 18]
011201156	[63, 92, 49, 78, 94, 39, 69, 22, 80, 86]
011201166	[94, 86, 83, 18, 30, 79, 34, 19, 71, 36]
011201170	[71, 53, 70, 44, 64, 83, 84, 39, 78, 98]
011201173	[12, 69, 41, 46, 58, 59, 22, 80, 87, 96]
011201176	[62, 90, 65, 80, 10, 27, 44, 28, 66, 16]
011201286	[26, 24, 98, 68, 62, 69, 91, 63, 56, 14]
011201334	[92, 36, 48, 40, 94, 62, 20, 93, 64, 90]
011201344	[34, 67, 44, 83, 40, 15, 59, 91, 51, 95]
011201352	[73, 45, 86, 32, 43, 67, 33, 98, 17, 87]
011201353	[51, 89, 67, 36, 41, 35, 10, 78, 75, 91]
011201366	[40, 17, 85, 15, 76, 31, 44, 65, 33, 19]
011201400	[53, 85, 71, 73, 90, 31, 97, 48, 12, 24]
011201403	[80, 46, 63, 76, 13, 92, 35, 74, 53, 34]
011201406	[80, 41, 76, 54, 64, 92, 96, 86, 98, 65]
011201446	[84, 11, 13, 65, 90, 36, 67, 92, 10, 63]
011202199	[63, 94, 97, 38, 37, 65, 46, 33, 47, 92]
Not registered	[29, 27, 74, 30, 33, 75, 48, 84, 49, 62]

